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RESEARCH ARTICLE

Validity and Reliability of a Mobile Device Application for Assessing Motor Performance in the 30-Second Sit-to-Stand Test

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ABSTRACT This study aimed to test the validity and reliability of a mobile device application for capturing center of mass (CoM) acceleration signals in analyzing and describing events during the 30-second chair stand test. A sample of 25 healthy young adults was evaluated using a smartphone with an Android-based application positioned at the fifth lumbar vertebra level (L5). Performance data from the mobile device were compared to those captured by a three-camera motion capture system, which served as the gold-standard reference. Signal synchronization between the smartphone and the kinematic system was achieved using an initial vertical jump, during which the peak in vertical acceleration from both systems was used as a temporal alignment marker. The motor behavior of the CoM was subsequently observed during task execution. Data were processed offline using routines developed for MatLab software, focusing on the longitudinal axis and describing the event cycles. A total of 21 variables were extracted for analysis. Each movement cycle was divided into three phases: impulse (DP1), sit-to-stand (DP2), and stand-to-sit (DP3). This approach allowed for the observation of cycle duration, the number of cycles performed, execution frequency, positive and negative acceleration areas (AUC+ and AUC-) for each phase, maximum and minimum acceleration peaks (Max Acc and Min Acc) per phase, and the amplitude between these peaks. Mean values for each variable were tested for sample normality, followed by criterion validation and reliability analyses. The hypothesis was relatively validated against the gold standard for 18 out of 21 variables in the smartphone app. Only the for P2 duration and accelerations in AUC+P1 and AUC+P2 showed correlation values < 0.70. Reliability was confirmed for 16 out of 21 variables; however, for DP1, DP2, and DP3 duration, as well as accelerations for Max Acc P2 and Min Acc P1, exhibited reliability values < 0.70. Despite some limitations, the mobile application demonstrated partial effectiveness in observing CoM motor behavior during the proposed functional test.

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INDEX TERMS Validation, accelerometry, inertial sensors, 30-second chair stand test, digital health, mHealth.

I. INTRODUCTION

The ability to sit and stand has been demonstrated as a key functional prerequisite for assessing mobility in various clinical conditions and in the context of population aging [1]. This is particularly relevant since transitioning from a seated to a standing position or vice versa is one of the most frequently performed activities throughout the day [2], [3], [4], [5]. These transitions inherently challenge dynamic balance, as they involve shifts in the center of mass and require continuous postural adjustments to maintain stability over a changing base of support. To evaluate this daily activity, the 30-Second Sit-to-Stand Test (30s STS) has been validated as part of a functional physical assessment battery and is particularly suited for capturing both strength and endurance within a time-constrained functional task, making it more representative of real-life repeated efforts than the 5-repetition STS, while being less fatiguing and more feasible than the 1-minute STS in healthy populations. [6]. Moreover, the 30s STS is widely used in clinical settings due to its efficiency, safety, and strong correlation with other functional measures, particularly in populations with chronic conditions. It also demonstrates sensitivity to rehabilitation outcomes, with an established minimal clinically important difference of two repetitions, supporting its use in monitoring treatment efficacy and functional change [7]. The goal of this test is to quantify the number of repetitions performed by the subject, providing insights into their lower limb strength and flexibility [8]. Although the 30-second Sit-to-Stand Test (30s STS) traditionally relies on counting the number of repetitions to estimate lower limb strength and endurance, this approach has limitations in capturing the complexity and quality of movement. In contrast, incorporating kinematic and kinetic variables offers clinicians and practitioners a more detailed view of motor performance. Kinematic parameters such as vertical acceleration, trunk inclination, and subphase durations allow the identification of subtle motor strategies that are often imperceptible when only repetition count is considered [9]. These variables have shown greater sensitivity in distinguishing levels of frailty in older adults and are associated with important clinical outcomes, such as recovery trajectories after orthopedic procedures [10]. By integrating these parameters into the 30s STS, assessments become not only more accurate but also more informative for designing individualized interventions [11].

To provide more robust data on performance in this function, researchers have explored the use of inertial sensors to capture acceleration data across the Cartesian axes (X-Y-Z) [12], [13]. These data points serve as observations for analyzing human movement patterns. With the development and sophistication of mobile devices, such as smartphones equipped with triaxial inertial sensors (accelerometers and

gyroscopes) [14], there has been growing interest in investigating their ability to provide data for analyzing body acceleration patterns and interpreting aspects such as movement direction, speed, force [15], and postural stability across different populations [16] or clinical conditions [17], [18].

One challenge lies in defining parameters that serve as references for movement recognition and analysis [19]. According to a systematic review [5], [20], there is still no consensus on the most effective analysis points for assessing motor performance in the 30s STS test. Nonetheless, researchers continue to work on describing movement patterns using acceleration data, isolating the sit-to-stand and stand-to-sit phases [5] or including the momentum phase to analyze initial trunk disturbances before transitioning from sitting to standing [10].

While some studies employ frequency-domain analysis parameters to interpret motion data [21], parameters derived from the time domain have been identified as more effective in detecting biomechanical variations in complex activities [22]. This distinction is particularly relevant in the context of smartphone-based applications, which are increasingly used to extract acceleration data for identifying patterns in human movement. Typically, such applications rely on analyzing all three axes of motion (X, Y, and Z) to characterize activities, ranging from static functions like sit-to-stand transitions [22] to dynamic tasks such as walking or climbing stairs [22], [23]. However, for the 30-Second Sit-to-Stand Test, focusing specifically on the Y-axis has demonstrated superior effectiveness in capturing event cycles, particularly for marking transitions such as leaving and returning to the chair [17], [20]. This approach enhances the precision of movement analysis, aligning with the test's objective of assessing lower limb functionality [24]. The selection of a young adult population reflects a deliberate methodological choice. This age group represents a physiological baseline, typically at the peak of neuromuscular performance, and is frequently used as a reference in functional validation studies. By minimizing the influence of age-related or pathological variability, this sample allows for more accurate determination of measurement sensitivity and reproducibility [25]. Moreover, understanding lower-limb performance in this population is relevant for applications in sport and exercise contexts, where early identification of performance deficits and monitoring of functional improvements are critical [27]. The integration of a mobile application for assessing 30-second sit-to-stand (30s STS) performance in young adults represents a meaningful advancement in the field of functional movement assessment. This approach leverages advanced sensor technology within a user-friendly and accessible platform capable of remotely monitoring biomechanical

events. By utilizing triaxial accelerometers and gravity sensors embedded in smartphones, the app enables precise cycle segmentation and captures the acceleration-based duration of each subphase within the sit-to-stand movement sequence. Previous studies have reported high reliability and validity for similar sensor-based systems, with intraclass correlation coefficients (ICCs) exceeding 0.90, thereby reinforcing the potential clinical and athletic applications of this technology [25]. Additionally, mobile apps like the one examined in this study promote scalable and remote data collection. This can contribute to personalized training programs, large-scale population screening, and early intervention strategies. It also empowers individuals to engage with their own physical health through self-assessment tools and enables clinicians to monitor patients outside of traditional clinical environments.

Thus, this research aimed to validate the use of a mobile device application, developed for the Android system, for analyzing and describing the event cycles of the 30-Second Sit-to-Stand Test. This was achieved by observing acceleration parameters of the CoM and verifying the reliability of measurements between test and retest. We hypothesize that the mobile application developed for the 30-Second Sit-to-Stand Test (30s STS) will demonstrate high test-retest reliability and measurement validity for detecting acceleration-based parameters of motor performance in healthy young adults. Furthermore, we expect that the app will enable precise cycle segmentation and subphase duration analysis, offering more comprehensive information on movement quality than repetition count alone.

II. MATERIALS AND METHODS

A. PARTICIPANTS

The study sample comprised 25 participants, including 13 women and 12 men, recruited through convenience sampling. Screening and assessments were conducted at the Laboratory of Human Movement of the Federal University of Pará. The sample size was calculated in BioEstat 5.3 (Belém, Pará, Brazil) using 80 % statistical power and 95 % confidence interval. Measurement was extracted from a pilot study performed in the first 7 subjects, in the first session. The mean and standard deviation for the Total duration (TD) was estimated as 1.554 ± 0.017 (s) in kinematic data and 1.556 ± 0.020 (s) in the Mobile data. A required sample size of 20 individuals per group was calculated. The mean age was 27 years (± 6.57), the average weight was 73 kg (± 13.62), and the mean height was 1.68 m (± 0.09). The Body Mass Index (BMI) averaged $25.69 (\pm 4.11)$. However, three participants withdrew from the study and did not participate in the retest, leaving a total of 22 participants for this phase (12 women and 10 men). For the retest group, the mean age was 27.5 years (± 6.57), the average weight decreased slightly to 71.85 kg (± 14.25), and the mean height remained consistent at 1.68 m (± 0.08). The BMI for the retest group averaged $25.52 (\pm 4.17)$. All participants were independent in activities of daily living, reported no functional limitations, and were

free from musculoskeletal pain or fatigue in the 30 days prior to testing. Additionally, none were under medication that could affect neuromuscular function or balance. Based on self-reports, most participants engaged in regular physical activity at least three times per week, including aerobic or resistance training. Participants were healthy young adults with no reported musculoskeletal conditions. Information on physical activity levels was also collected to contextualize neuromuscular performance baselines. None of the participants were professional athletes, and all reported engaging in resistance exercise approximately three times per week. These criteria ensured that the sample represented a healthy, physically active adult population at their neuromuscular performance peak. Participants were excluded if they had a history of orthopedic, neurological, or rheumatic conditions, or any other health issues that could potentially interfere with task performance. Ethical approval for the study was obtained from the local ethics committee prior to its initiation (Approval Report #3.966.255).

B. EQUIPMENT

An Android smartphone (Samsung Galaxy A10 model, weight: 170g, dimensions: $15.5 \times 7 \times 0.07$ cm), equipped with accelerometer sensors and Octa-Core processors (2 GHz and 1.5 GHz), was used in this study. Accelerometry data were remotely captured and exported through a custom-developed Android application coded (Momentum App, with a sampling rate of approximately 63 Hz) installed on the smartphone.

The Momentum app was developed by our research group using the Java programming language for Android operating systems. It is currently in its beta version and not yet commercially available. The application's underlying code utilized the Android SDK library, which provides functions for accessing sensors embedded in mobile devices. The application's output was designed to report oscillations across three axes: vertical (Y), anteroposterior (Z), and mediolateral (X).

As a gold-standard reference method for motion analysis, three kinematic capture cameras (Simi Motion, Simi, Germany) were employed. Since STS movements involve low-frequency components (typically below 10 Hz), a sampling frequency of 120 Hz was used to ensure accurate temporal resolution [10], [20]. Data from the cameras were transmitted to a computer at the Laboratory via USB cables and analyzed using proprietary software provided with the cameras.

C. EXPERIMENTAL PROCEDURE

Participants stood barefoot with their feet shoulder-width apart. A mobile device was secured at the level of the fifth lumbar vertebra (L5) using a belt around the waist, and a reflective marker was affixed directly onto the device for motion tracking. To minimize motion artifacts, the smartphone was firmly secured at the L5 level using a neoprene belt with moderate compression, ensuring stable contact with the trunk. This anatomical site, close to the body's center

of mass in the sagittal plane, offers minimal soft tissue displacement and has been validated in previous studies as a reliable position for capturing trunk acceleration and segmenting sit-to-stand cycles [10], [28]. All data were collected simultaneously (Figure 1). Given that the participants in this study were young, healthy, and physically active adults, they were first instructed to perform a vertical jump while maintaining the same position, to synchronize the signals from both assessment instruments. Signal alignment was achieved by identifying the peak on the vertical axis, which corresponded to the moment of ground impact. After this stage, participants were evaluated using the 30-Second Sit-to-Stand Test (Figure 1). In this test, the participant was required to sit and stand from a wooden bench (seat height: 43 cm, without armrests or back support) as many times as possible within 30 seconds, following the procedure described by Jones et al. [8]. To ensure consistency in joint positioning, all participants began the test seated upright on the wooden bench (43 cm in height), with their backs unsupported, hips positioned near the edge of the seat, and knees flexed at approximately 90 degrees with feet flat on the ground, shoulder-width apart. This posture aimed to minimize variability in hip flexion at the onset of the ascent phase and standardize the initial conditions for accelerometry data collection. They were instructed to look straight forward and to rise after the “1, 2, 3, go” command at their own preferred speed with their arms folded across their chest. All trials were performed using the same chair and with similar ambient conditions. The evaluation was conducted in two sessions (test and retest). In each phase, participants performed two initial trials to familiarize themselves with the activity, and during the third trial, acceleration data from the CoM were collected. The retest was performed one week later to determine the reliability of the analytical parameters.

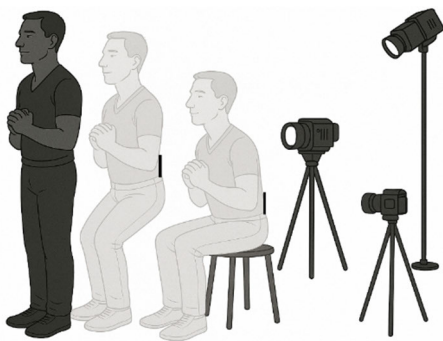


FIGURE 1. Experimental protocol: with the subject with his arms crossed, sitting on the bench; smartphone inside an adapted belt and the photoluminescent marker on the belt, positioned on the fifth lumbar vertebra. The individual gets up and sits the most times in 30 seconds.

D. SIGNAL PROCESSING AND MOVEMENT PHASE SEGMENTATION

Synchronization, processing, and data analysis were performed offline using MATLAB R2020 software (MathWorks,

Natick, MA, USA), through a specific script for segmenting event cycles, divided into two stages. The acceleration data was filtered to remove noise using a 2th-order Butterworth low-pass 5 Hz filter and eliminating the gravitational component of the smartphone for subsequent comparison with kinematic data. To analyze the sit-to-stand transitions, we employed the vertical acceleration signal obtained from the smartphone's triaxial accelerometer. This signal was processed using a custom script to segment complete sit-stand-sit cycles based on distinctive features in the waveform. The beginning of the first sit-to-stand cycle was identified by a prominent peak in vertical acceleration (Y-axis), corresponding to the seat-off moment, when the participant initiated trunk propulsion. Crucially, all acceleration-based event identifications were visually confirmed using synchronized kinematic data from the motion capture system, which served as the gold-standard reference. This visual verification ensured precise cross-validation of the seat-off, upright, and reseating events throughout the movement cycle. Each full cycle (sit-stand-sit) was delineated by identifying the vertical acceleration peaks (representing standing transitions) and troughs (indicating reseating moments). The stand-up phase was defined as the interval from seat-off to the point of full leg extension and upright trunk posture, typically marked by a secondary peak. The sitting phase ended when the participant fully reloaded the chair, detected as a local minimum followed by signal stabilization in the vertical acceleration curve. We intentionally avoided using the first point of chair contact as the endpoint of the sitting phase, since this can occur prior to complete weight transfer. Instead, we relied on trunk stabilization and cessation of downward acceleration, as confirmed by the motion capture recordings, to define the end of the cycle. This segmentation method was consistently applied to all repetitions in the 30-second Sit-to-Stand Test, providing reliable and standardized detection of movement subphases across participants. Figure 2 illustrates the correspondence between the acceleration signal and the movement events used for cycle segmentation, as verified through kinematic analysis.

Once the 30-s signals were cut into sit-stand-sit cycles, each individual cycle was divided into different movement phases (i.e. impulse, stand-up and sit-down). During the impulse phase the subject accommodates its weight to the chair and obtains the necessary push to leave the chair again and reach the upright position. The stand-up phase starts when the subject leaves the seat and finishes when he completely stands-up. And, finally, the sit-down phase comprises the movement needed to reach the chair from the previous upright position. Throughout the text the boundaries between the impulse/stand-up, stand-up/sit-down and sit-down/impulse phases will be referred to as P1, P2, and P3 respectively (Figure 3). The events that define the phases' boundaries were detected from distinct marks in the kinematics visual inspection and so they were analyzed to assess the test performance. To support the segmentation criteria adopted in this study, we relied on evidence from prior

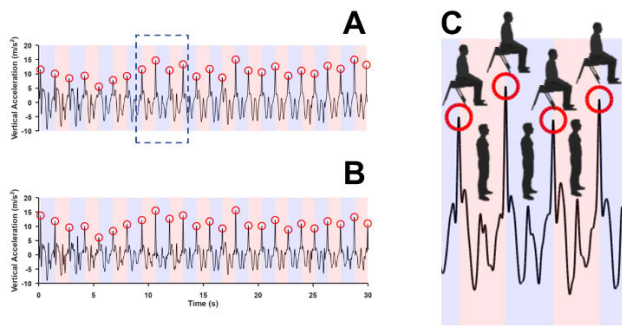


FIGURE 2. Vertical acceleration signals obtained from the kinematic system (A) and the Momentum App (B) during the 30-second Sit-to-Stand Test. Red circles indicate the key time points used to segment the sit-stand-sit cycles, corresponding to the moments of seat-off (initiation of standing) and full re-seating (end of sitting). Panel (C) provides a visual representation of the signal features associated with these events, illustrating how the cycle segmentation was based on peaks (standing) and troughs (seating) in the vertical acceleration curve.

research that validates the use of vertical acceleration patterns to delineate sit-to-stand events [10], [20], [29].

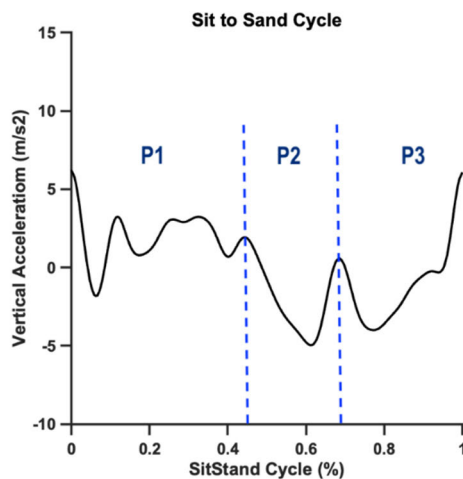


FIGURE 3. Sit-stand-sit cycle diagram with phases on Z-acceleration signal. The dashed blue lines show respectively the P1, P2 and P3 division.

21 variables were extracted (test and retest) for each participant (Figure 4):

- NC: Number of completed cycles
- F: Execution Frequency of each sit-to-stand cycle (in Hz)
- TD: Total Duration of each STS cycle (in seconds)
- DP1, DP2 and DP3: Duration of each phase (in sec)
- Max Acc P1, Max Acc P2, and Max Acc P3: Maximum acceleration of each phase (m/s^2)
- Min Acc P1, Min Acc P2, and Min Acc P3: Minimum acceleration of each phase (m/s^2)
- Range P1, Range P2 and Range P3: Range of each phase (m/s^2)
- AUC+P1, AUC+P2 and AUC+P3: Positive component of the acceleration's area under the curve related to the movement duration ($\text{m/s}^2/\text{s}$)

- AUC-P1, AUC-P2 and AUC-P3: Negative component of the acceleration's area under the curve related to the movement duration ($\text{m/s}^2/\text{s}$)

The AUC was defined as the acceleration's area under the curve related to the movement duration. This parameter expresses the sum of the acceleration amplitudes in order to take into account not only amplitudes but also time length. According to the displacement's direction the AUCacc is composed of a positive component (AUC+acc, with the movement's direction) and a negative one (AUC-acc, with the opposite direction). The first parameter evaluates the necessary impulse to carry out each transition (movement acceleration); while the second one is related to the transition control since the first impulse must be compensated to reach the stand-up or the sit-down position (movement deceleration).

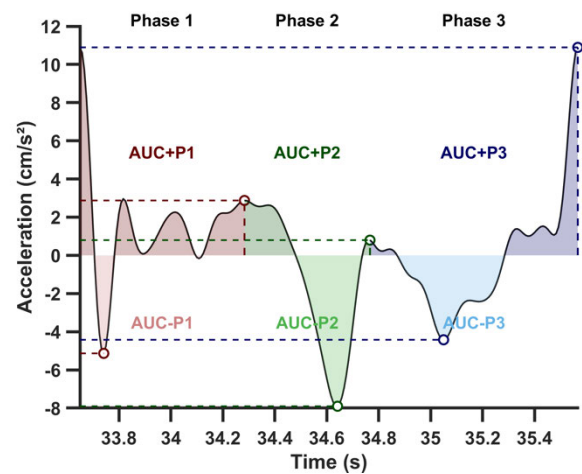


FIGURE 4. Definition of the movement cycle phases in the Sit and Stand phases, with their dashed lines on the x-axis, dividing the area of the impulse phase cycles (P1) in red; sit-stand phase (P2) in green area and stand-up phase (P3) in blue area. It also presents the observation criteria in each phase, with its areas of positive acceleration (AUC+) and negative acceleration (AUC-). With its maximum peaks (Max Acc) and minimums (Min Acc) demonstrated by the dashed lines in the y-axis.

E. STATISTICAL ANALYSIS

The RStudio software was used for data analysis. Initially, the normality of the distribution for each variable was assessed using the Shapiro-Wilk test. For variables that followed a normal distribution, means and standard deviations were reported; for non-normally distributed variables, medians and interquartile ranges were presented. To evaluate criterion validity, concurrent validation was performed using Pearson's correlation coefficient (r) for parametric variables and Spearman's correlation coefficient (ρ) for non-parametric variables. Correlation coefficients were interpreted following Abbott et al. [30]. Subsequently, Bland-Altman plots were constructed with 95% limits of agreement to compare the mean values obtained from the devices. In the Bland-Altman analysis, we computed the mean bias (i.e., the average difference between devices), the limits of agreement (defined

as the bias $\pm 1.96 \times$ the standard deviation of the differences), and the agreement range (calculated as the difference between the upper and lower limits of agreement). A bias was considered statistically significant when zero fell outside the 95% confidence interval of the mean bias. Relative reliability and systematic bias between devices were assessed using the Intraclass Correlation Coefficient (ICC), with interpretation criteria based on Shrout and Fleiss [31]. For both criterion validity and reliability, acceptable thresholds were set at $r/\rho \geq 0.70$ and $ICC \geq 0.70$, as recommended by Souza et al. (2017). To strengthen the assessment of measurement properties, absolute reliability indices were also calculated, for parametric data, the Standard Error of Measurement (SEM), Coefficient of Variation (CV), Minimal Detectable Change at 95% confidence (MDC95), Absolute Percent Error (APE), and accuracy, while for non-parametric data, we calculated equivalent parameters including non-parametric ICC in which is calculated the value of $1 - \frac{IQR_{difference}}{IQR_{test} + IQR_{retest}}$, Standard Error of Median via bootstrap sampling ($n = 1000$), Quartile Coefficient of Variation (CVIQR which is the ratio between interquartile range and sum of first and third quartiles), non-parametric Minimal Detectable Change via bootstrap ($n = 1000$), Median of the Absolute Percent Errors. These indices allow for a comprehensive evaluation of test-retest agreement and error margins, supporting clinical and practical interpretations of the results.

III. RESULTS

Of the 21 analyzed variables, 18 demonstrated relative validation against the gold standard of this study (Table 1). The correlation analysis between the data captured by the evaluation instruments revealed near-perfect indices for temporal variables in parametric data ($r = 0.944$ – 0.999) and moderate to near-perfect indices in non-parametric data ($\rho = 0.408$ – 1). The acceleration parameter variables showed strong to near-perfect correlations for parametric data ($r = 0.698$ – 0.974) and non-parametric data ($\rho = 0.685$ – 0.963).

In the supplementary figure the Bland-Altman plots (see supplementary file 1: 18 plots A to R), the value zero falls within the 95% confidence limits, suggesting that there is no significant systematic bias between the compared methods. This indicates good agreement across all analyzed parameters.

Table 2 presents the reliability outcomes for test-retest comparisons using both the kinematic system and the mobile application across temporal and acceleration-related variables. High intraclass correlation coefficients ($ICC > 0.80$) were observed for total duration (TD) and certain acceleration metrics (e.g., AUC+ P1, Min Acc P2), indicating good to excellent reliability. The Momentum app showed comparable or superior performance to the kinematic reference in several measures, particularly in AUC+ P1 and Min Acc P2, where it reached ICCs of 0.85 and 0.91, respectively. Standard Error of Measurement (SEM), Minimal Detectable Change (MDC), and Absolute Percent Error (APE) values further support the

TABLE 1. Result of variance tests in the criterion validation analysis of the App. Momentum.

Variable	Kinem	Mobile	<i>r</i> or <i>rho</i>	<i>p</i> -value
TD (sec)	1.55 (0.22)	1.55 (0.22)	$r = 0.99$	< 0.000
DP1 (sec)	0.56 (0.25)	0.57 (0.22)	$r = 0.97$	< 0.000
DP2 (sec)	0.41 (0.39 / 0.43)	0.39 (0.38 / 0.43)	$\rho = 0.40$	0.043
DP3 (sec)	0.57 (0.07)	0.57 (0.07)	$r = 0.94$	< 0.000
NC (n°)	19.00 (18.00 / 20.00)	19.00 (18.00 / 20.00)	$\rho = 1$	< 0.000
F (Hz)	0.63 (0.60 / 0.66)	0.63 (0.60 / 0.66)	$\rho = 1$	< 0.000
AUC+ P1 (m/s ² /s)	1.21 (0.24)	1.41 (0.37)	$r = 0.698$	0.000
AUC+ P2 (m/s ² /s)	0.32 (0.22 / 0.40)	0.21 (1.13 / 0.26)	$\rho = 0.685$	0.000
AUC+ P3 (m/s ² /s)	0.73 (0.51 / 0.86)	0.60 (0.45 / 0.79)	$\rho = 0.713$	< 0.000
AUC- P1 (m/s ² /s)	0.16 (0.08 / 0.32)	0.13 (0.10 / 0.39)	$\rho = 0.881$	< 0.000
AUC- P2 (m/s ² /s)	1.09 (0.88 / 1.30)	0.98 (0.84 / 1.33)	$\rho = 0.870$	< 0.000
AUC- P3 (m/s ² /s)	0.93 (0.20)	0.97 (0.23)	$r = 0.901$	< 0.000
Max Acc	9.28	11.80	$\rho = 0.953$	< 0.000
P1 (m/s ²)	(7.88 / 11.59)	(9.14 / 13.45)	$\rho = 0.943$	< 0.000
Max Acc	4.17	3.71	$\rho = 0.941$	< 0.000
P2 (m/s ²)	(3.48 / 4.99)	(2.84 / 5.81)	$\rho = 0.941$	< 0.000
Max Acc	9.37	12.10	$\rho = 0.974$	< 0.000
P3 (m/s ²)	(7.88 / 11.59)	(9.19 / 14.30)	$\rho = 0.898$	< 0.000
Min Acc	-3.24	-3.69	$r = 0.974$	< 0.000
P1 (m/s ²)	(2.32)	(2.62)	$\rho = 0.917$	< 0.000
Min Acc	-6.74	-6.55	$\rho = 0.963$	< 0.000
P2 (m/s ²)	(1.86)	(1.99)	$\rho = 0.963$	< 0.000
Min Acc	-5.45	-5.18	$\rho = 0.957$	< 0.000
P3 (m/s ²)	(-6.36 / -4.52)	(-6.49 / -4.58)	$\rho = 0.957$	< 0.000
Range P1 (m/s ²)	11.78 (9.98 / 16.22)	15.35 (10.90 / 18.08)	$\rho = 0.96$	< 0.000
Range P2 (m/s ²)	10.26 (8.55 / 13.32)	10.10 (7.90 / 14.18)	$\rho = 0.963$	< 0.000
Range P3 (m/s ²)	15.41 (12.83 / 19.71)	17.36 (13.98 / 21.53)	$\rho = 0.957$	< 0.000

Note. Analysis variables for mobile and Kinematics (Kinem): duration of the total cycle and impulse phases (DP1); sit/stand (DP2); stand/sit (DP3); number of cycles (NC) and frequency (F) of execution. As well as acceleration variables: positive acceleration (AUC+) and negative (AUC-) areas; maximum (Max Acc) and minimum (Min Acc) acceleration peaks and their amplitudes per phase. Presentation of means and standard deviation (parametric), median and interquartile amplitude (non-parametric); as well as Pearson correlation (*r*) in results parametric; Spearman correlation (*Rho*) in nonparametric. In bold, statistically significant results.

accuracy and consistency of the app, especially in early movement phases. However, lower reliability was noted for later phases (DP2, DP3) and some range parameters, suggesting greater variability in movement execution or detection.

IV. DISCUSSION

The present study aimed to validate the Momentum App, an Android-based application designed to extract acceleration-derived parameters during the 30-second Sit-to-Stand Test (30s-STs) in healthy young adults. Our results provide support for the app's ability to segment and quantify the sit-stand-sit cycles based on vertical acceleration, enabling detailed analysis of subphase durations and magnitudes. This discussion is structured around three core themes: 1) validity and reliability of the app, (2) clinical relevance

TABLE 2. Result of the reliability test between the averages of the test and retest by instrument in parametric and nonparametric variables.

Variable	Instrument	ICC (law / Upp limit's) or ICC (Ranks)	p-value	SEM	CV% or IQR/median		MDC	APE (%)	Accuracy (%)
					Test	Retest			
TD (s)	Kinem	0.80 (0.53/0.91)	< 0.000	0.22	14.86	14.74	0.62	9.49	89.92
	Mobile	0.81 (0.56 / 0.92)	< 0.000	0.22	14.91	14.69	0.62	9.25	90.19
DP1 (s)	Kinem	0.82 (0.58/0.92)	< 0.000	0.23	46.02	43.24	0.66	23.31	74.18
	Mobile	0.70	---	0.059	43.74	55.76	1.20	---	---
DP2 (s)	Kinem	0.47	---	0.007	10.61	22.29	0.26	---	---
	Mobile	0.26	---	0.010	13.25	15.85	1.13	---	---
DP3 (s)	Kinem	0.46 (-0.28/0.77)	0.079	0.11	12.45	27.58	0.32	16.80	82.73
	Mobile	0.48 (-0.22 / 0.78)	0.065	0.12	12.92	29.61	0.35	17.62	82.11
NC (n)	Kinem	0.57	---	0.57	10.53	25.00	7.43	---	---
	Mobile	0.57	---	0.59	10.53	25.00	7.48	---	---
F (Hz)	Kinem	0.57	---	0.01	10.53	25.00	0.25	---	---
	Mobile	0.57	---	0.01	10.53	25.00	0.24	---	---
AUC+ P1 (m/s ² /s)	Kinem	0.68 (0.25 / 0.86)	0.004	0.23	20.46	20.76	0.66	13.18	85.67
	Mobile	0.85 (0.65 / 0.94)	0.000	0.40	27.45	33.06	1.13	15.93	82.91
AUC+ P2 (m/s ² /s)	Kinem	0.66	---	0.01	59.41	39.62	0.44	---	---
	Mobile	0.55	---	0.01	58.58	55.14	0.38	---	---
AUC+ P3 (m/s ² /s)	Kinem	0.68 (0.25 / 0.86)	0.004	0.23	35.73	27.38	0.65	20.28	78.31
	Mobile	0.79	---	0.05	47.64	47.32	0.57	---	---
AUC- P1 (m/s ² /s)	Kinem	0.47	---	0.03	122.70	152.26	0.38	---	---
	Mobile	0.62	---	0.03	167.71	140.45	0.34	---	---
AUC- P2 (m/s ² /s)	Kinem	0.81	---	0.04	36.06	23.62	0.78	---	---
	Mobile	0.81	---	0.05	49.93	32.55	0.67	---	---
AUC- P3 (m/s ² /s)	Kinem	0.31 (-0.61 / 0.71)	0.189	0.27	22.61	37.59	0.77	22.24	77.16
	Mobile	0.76	---	0.03	38.58	25.73	0.76	---	---
Max Acc P1 (m/s ²)	Kinem	0.56	---	0.59	39.89	72.00	9.21	---	---
	Mobile	0.50	---	0.67	35.31	67.36	13.36	---	---
Max Acc P2 (m/s ²)	Kinem	0.68	---	0.16	43.44	37.76	3.91	---	---
	Mobile	0.71	---	0.42	88.94	67.06	6.09	---	---
Max Acc P3 (m/s ²)	Kinem	0.54	---	0.53	36.80	71.54	9.45	---	---
	Mobile	0.76 (0.43 / 0.90)	0.000	6.08	40.72	50.30	16.85	34.80	68.40
Min Acc P1 (m/s ²)	Kinem	0.53	---	0.34	-95.54	-103.87	5.10	---	---
	Mobile	0.53	---	0.40	-98.75	-88.58	5.81	---	---
Min Acc P2 (m/s ²)	Kinem	0.87 (0.70 / 0.94)	0.000	1.99	-18.12	-30.87	5.53	15.59	113.87
	Mobile	0.91 (0.80 / 0.96)	0.000	1.94	-29.90	-29.37	5.39	13.30	112.23
	Kinem	0.69	---	0.36	-38.55	-47.66	7.33	---	---

TABLE 2. (Continued.) Result of the reliability test between the averages of the test and retest by instrument in parametric and nonparametric variables.

Min Acc P3 (m/s ²)	Mobile	0.70	---	0.34	-35.19	-40.30	5.24	---	---
Range P1 (m/s ²)	Kinem	0.69	1.94		-29.90	-29.37	5.39	13.30	112.23
	Mobile	0.69	---	1.39	44.26	82.62	16.19	---	---
Range P2 (m/s ²)	Kinem	0.86 (0.66 / 0.94)	0.000	3.22	3.18	27.71	8.93	14.25	86.11
	Mobile	0.72	---	0.68	58.91	44.41	7.19	---	---
Range P3 (m/s ²)	Kinem	0.60	---	0.69	43.20	42.62	11.52	---	---
	Mobile	0.57	---	0.88	42.68	48.74	15.28	---	---

Note. Reliability test of parametric measurements by instrument for temporal variables: total duration (DT); impulse (DP1); sit/decelerate (DP2); stand/sit (DP3); cycles and frequency. As well as acceleration variables: areas of positive (AUC+) and negative (AUC-) acceleration; peaks of maximum (Acc Max.) and minimum (Acc Min.) acceleration and their amplitudes per phase. Presentation of mean and standard deviation data, Standard Error of Measurement (SEM), Coefficient of Variation (CV%), Minimal Detectable Change (MDC), Absolute Percent Error (APE), Accuracy (% agreement), Intraclass Correlation Coefficient (ICC) for the evaluation instruments in test and retest.

and practical applications, and (3) implications for future research.

The analysis focused on the observation of the ‘Y’ axis (longitudinal on the device) to monitor the phases of standing up and returning to the chair, as previously evaluated in other studies [17], [29], [32].

A. VALIDITY AND RELIABILITY OF THE MOMENTUM APP

The validity results observed for the *Momentum App* demonstrate a high degree of concordance with the kinematic gold standard, particularly in temporal and acceleration-related parameters. Nearly all variables exhibited strong to near-perfect correlations (e.g., TD: $r = 0.99$, DP1: $r = 0.97$, Min Acc P2: $r = 0.974$), indicating robust relative validity. These findings are consistent with previous literature. For example, Boonstra et al. reported excellent agreement between inertial sensors and optical motion capture systems for temporal features of sit-to-stand movements [33]. Similarly, Doheny et al. found that accelerometer-based measures such as spectral edge frequency and movement jerk correlated strongly with faller status and standard functional measures, supporting the validity of inertial data in clinical contexts [34]. In the present study, the app also performed well across most acceleration-derived variables, with rho values above 0.85 for Max Acc and Range parameters across movement phases, aligning with the high correlations reported by Millor et al. in their biomechanical analyses of sit-to-stand performance [20]. Notably, while DP2 showed only moderate correlation ($\rho = 0.40$), this is consistent with prior observations of greater variability and segmentation challenges in mid-phase transitions [20].

The test-retest reliability outcomes of the *Momentum App* align with and in some cases exceed values reported in previous studies utilizing inertial sensors for sit-to-stand assessments. Consistent with Doheny et al. (2011), who reported ICCs ranging from 0.71 to 0.92 for temporal and acceleration variables during the Five Times Sit-to-Stand Test (FTSS), the current study observed good to excellent ICC values (≥ 0.80) for total duration and specific acceleration metrics such as Min Acc P2 (ICC = 0.91) and AUC+ P1

(ICC = 0.85) [35]. These results support the robustness of the app, particularly in the initial phases of movement execution, as also highlighted by Millor et al., who emphasized the biomechanical importance of the impulse phase [10]. Notably, the reliability for the mobile instrument was comparable or superior to that of the reference kinematic system in key parameters. However, as observed in prior studies (e.g., Millor et al., 2014; Boonstra et al., 2006), later phases of the movement (DP2, DP3) and composite measures such as acceleration range demonstrated lower ICCs (e.g., DP2: 0.26; Range P3: 0.57), possibly reflecting higher intra-individual variability and technical challenges in segmenting complex motor events [20], [33]. Despite this, the small Standard Error of Measurement (SEM) and favorable Absolute Percent Error (APE) values across most parameters further reinforce the accuracy and consistency of the *Momentum App* for quantifying sit-to-stand performance.

B. CLINICAL RELEVANCE AND PRACTICAL APPLICATIONS

This study reinforces the clinical relevance of using inertial sensor-based mobile applications, such as the *Momentum App*, for the quantitative assessment of functional performance in healthy young adults. Among the various parameters analyzed, temporal variables, especially total duration (TD) and early phase duration (DP1), stand out as the most clinically informative due to their high reliability and ability to reflect neuromuscular coordination. Importantly, acceleration-derived parameters such as peak values and the area under the curve (AUC $\pm$), particularly during the impulse phase (P1), have also demonstrated high clinical sensitivity, consistent with findings in previous research [20], [33]. While Jones et al. originally proposed the 30s-STs as a practical measure based on repetition count [8], more recent studies, such as those by Doheny et al., have demonstrated that accelerometry-derived parameters, such as spectral edge frequency and acceleration variability, offer clinically relevant insights, including the ability to distinguish fallers from non-fallers [34], [35]. This shift from quantity to quality in movement analysis is echoed in our findings, which demonstrate that the *Momentum App* can effectively identify

fine-grained temporal and dynamic markers, even in high-functioning populations.

Additional studies have shown that IMUs embedded in everyday devices, like smartphones or smart glasses, offer precision comparable to laboratory systems in functional tests such as the 5 Times Sit-to-Stand (5xSTS), with RMSE values as low as 3.26% and ICCs ranging from 0.78 to 0.91. [36] [37]. Our results align with this growing body of evidence, confirming that mobile IMU-based tools can provide high-resolution motion data in real-time, outside controlled lab environments. This real-world accessibility has direct implications for clinical monitoring, workplace screening, and even remote health programs.

Recognizing patterns in sit-to-stand transitions has also proven clinically valuable in contexts such as frailty detection, fall-risk prediction in multiple sclerosis, and motor symptom tracking in Parkinson's disease [38]. In particular, prolonged execution phases have been inversely associated with muscle power output and whole-body functional capacity, highlighting the utility of phase-specific timing as a diagnostic marker [39], [40]. Our study supports this approach by demonstrating that certain sit-to-stand phases, especially the initial movement impulse (P1), carry biomechanical and functional relevance, and may serve as early indicators of motor decline or inefficiency, even in non-clinical populations.

Thus, recognizing sit-to-stand patterns offers clinical value for identifying frailty, predicting fall risk, and monitoring motor symptoms in neurological conditions. Phase-specific timing has been associated with muscle power and functional capacity, reinforcing its use as a diagnostic marker. The *Momentum App* enables accurate, sensor-based analysis in real-world settings, allowing early detection of motor inefficiencies and asymmetries, even in high-functioning individuals. For clinicians, this mobile technology expands access to objective, detailed movement data, supporting preventive care and performance optimization in populations often overlooked by traditional assessments. Overall, the *Momentum App* contributes to bridging the gap between research-grade assessments and real-world application, offering a practical yet scientifically grounded resource to support data-informed decisions in clinical, athletic, and preventive health contexts.

C. IMPLICATIONS FOR FUTURE RESEARCH

Overall, the *Momentum App* contributes to bridging the gap between research-grade assessments and real-world application, offering a practical yet scientifically grounded resource to support data-informed decisions in clinical, athletic, and preventive health contexts. Future research is essential to expand the clinical applicability of mobile-based assessments like the *Momentum App*, particularly in diverse populations and clinical contexts. While current findings demonstrate strong validity and reliability in healthy young adults, extending this approach to older adults or individuals

with neurological or musculoskeletal impairments could reveal important diagnostic and monitoring benefits. Additionally, incorporating automated segmentation algorithms and real-time feedback mechanisms may improve data processing efficiency and user experience, enabling broader implementation in clinical, home-based, or telehealth settings. By addressing current methodological limitations, such as manual event marking or variability in sampling rates, future studies can enhance the precision, scalability, and interpretability of digital mobility assessments, ultimately contributing to more personalized and proactive healthcare strategies.

DATA AVAILABILITY

Bland-Altman plots and all data that support the findings of this study are included within this paper and its Supplementary Information files.

COMPETING INTERESTS

The author(s) declare no competing interests.

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